SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:40**

Annexure A

CLASS TEST

Code of Conduct:

I ________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	**/ 30**	**/ 20**	**/ 15**	**/ 10**	**/ 100**

CO4 - ASSESSMENT - 1

Compiler Design

class Test

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38/40

① given:

if ((a<b)) and ((c<d) or (a>d)) then

z = x+y*z

else

z = z+1

Tree Address code

L1:

if a<b goto L2

goto L5

L2:

if c<d goto L4

goto L3

L3: if a>b goto L4

goto L5

L4: t1 = y*z

$t_2 = x + t_1$

$x = t_2$

goto L6

L5:

$t_3 = z+1$

$z = t_3$

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(2). Translate (a < b) and (c > d) or (e < f) into three address element.

Backpatch contion (a<b) goto 3

if a<b goto 3

goto 5

if c < d goto 7

goto 5

if e<f goto 7

goto 8

TRUE

FALSE

(3) consider the arithmetic expression

A = (-A+B)*(-A+B)*C

t1 = -A

t2 = t1 + B

t3 = -A

t4 = t3 + B

t5 = t4 + C

t6 = t2 + t5

A = t6

Quad ruples :

NO	operator	Arg1	Arg2	Result
t1	uminus	A	-	t1
t2	+	t1	B	t2
t3	uminus	A	-	t3
t4	+	t3	B	t4
t5	*	t4	c	t5
t6	*	t2	t5	t6
	=	t6	-	A

④ construct three address code using the following production for boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 \,||\, B_2$

$B \rightarrow B_1 \,\&\&\, B_2$

$B \rightarrow B_1$

$B \rightarrow E$ relop E_2

$B \rightarrow$ True

if a>b goto L True
goto L2

L2:

if c<d goto L TRUE
goto L3

L3:

if a < ~~d~~ goto L TRUE
goto L FALSE

L TRUE:

// Boolean expression is true

goto L END

L FALSE:

// Boolean expression is False.

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconception s	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps